

The sampling method used for the subsequent research work was representative sampling, to ensure that only those participants targeted truly represent the larger population, which, as a result, helps improve the overall accuracy of the research work to be carried out.

20.3.2 RESEARCH INSTRUMENT

A quantitative approach was employed to gather the participants' responses, involving the use of a matrix-based survey questionnaire, based on a five-point Likert scale, ranging from "strongly agree" to "strongly disagree." The survey questionnaire was designed and then divided into two main segmentations: (1) demographic segmentation and (2) attitudinal and behavioral segmentation. The demographic segmentation part of the survey questionnaire was based on respondents' demographic profiles, comprising four items: (1) gender, (2) age, (3) level of education, and (4) work experience.

Table 20.1 shows respondents' demographic profiles in terms of their gender, age, level of education, and work experience. The attitudinal and behavioral

TABLE 20.1

The Demographic Profiles of the Respondents (N 469)

Variables		Frequencies	Percentages (%)
(1) Gender	Male	316	67.4
	Female	153	32.6
	Total	469	100
(2) Age	18–25 Years	128	27.3
	26–35 Years	191	40.7
	36–45 Years	107	22.8
	46–55 Years	43	9.2
	56 Years or Above	0	0
	Total	469	100
(3) Level of Education	Diploma	7	1.5
	BA/BSc/BBA	290	61.8
	MA/MSc/MBA	145	30.9
	MPhil	27	5.8
	PhD	0	0
	Total	469	100
(4) Work Experience	1–2 Years	128	27.3
	3–5 Years	85	18.1
	6–8 Years	58	12.4
	9–12 Years	71	15.1
	13–16 Years	60	12.8
	17–20 Years	32	6.8
	21 Years or Above	35	7.5
	Total	469	100

Source: Author's owner.

Table 20.1 shows quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality and employee performance.

segmentation part of the survey questionnaire was based on the attitudinal and behavioral research items – based on the key independent and dependent variables of this study, i.e., quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality and employee performance.

20.4 RESULTS AND DISCUSSION

20.4.1 RESULTS

Reliability and validity analysis IBM SPSS Statistics was used for the data analysis. The reliability of the research instrument was tested using Cronbach’s alpha, whereas the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett’s test of sphericity were used to test the research instrument’s validity. Table 20.2 shows the results of the reliability and validity analysis.

The existing literature on Cronbach’s alpha shows that to pass the reliability test, each variable of the research instrument needs to have a Cronbach’s alpha value greater than or equal to 0.7.

Table 20.2 shows that each variable of the research instrument had a Cronbach’s alpha value greater than 0.7. Therefore, the reliability test was successfully passed by the research instrument. To pass the validity test, it is generally agreed that each variable of the research instrument should have a KMO value greater than or equal to 0.5.

Furthermore, each variable of the research instrument should also have a p-value (Sig.) of less than 0.05. Table 20.2 shows that each variable of the research instrument had a KMO value greater than 0.5.

Additionally, each variable of the research instrument had a p-value of 0.000. The validity test was accordingly also successfully passed by the research instrument. It can therefore be concluded that the research instrument was proven to be both reliable and valid.

TABLE 20.2
Reliability and Validity Analysis

Variable	No. of Items	Cronbach’s Alpha (α)	KMO Measure of Sampling	Bartlett’s Test of Sphericity Approx.		
			Adequacy	Chi-Square	df	Sig
Quick & Efficient Response	5	0.930	0.883	1888.292	10	0.000
Safe & Simple Working	8	0.951	0.947	3598.760	28	0.000
Security & Trust	7	0.943	0.931	3187.108	21	0.000
Timely Completion of Work	7	0.950	0.945	3475.500	21	0.000
Work Quality	6	0.935	0.922	2529.185	15	0.000
Employees’ Performance	5	0.953	0.899	2806.951	10	0.000

Source: Author’s owner.

TABLE 20.3**Correlations among All Variables of the Research Instrument**

Correlations	Quick and Efficient Response	Safe and Simple Working	Security and Trust	Timely Completion of Work	Work Quality	Employees' Performance
Quick & Efficient Response	1					
Safe & Simple Working	0.642**	1				
Security & Trust	0.529**	0.669**	1			
Timely Completion of Work	0.472**	0.567**	0.708**	1		
Work Quality	0.527**	0.471**	0.518**	0.611**	1	
Employees' Performance	0.421**	0.433**	0.438**	0.427**	0.489**	1

Note(s): **Correlation is significant at the 0.01 level (2-tailed).

Source: Author's owner.

20.4.2 CORRELATION AND REGRESSION ANALYSIS

A correlation analysis was carried out to evaluate the co-relationship among all the variables of the research instrument. Table 20.3 shows the results of the correlation analysis, which are positive and statistically significant for all the variables of the research instrument: quick and efficient response, safe and simple working, security and trust, timely completion of work, work quality, and organizational performance. Table 20.4 shows the correlation analysis results between Industry 4.0 (a single variable computed by adding the five variables of Industry 4.0) and employees' performance, as in Figure 20.2.

All variables of the research instrument were positive and significantly correlated with each other. Therefore, the obtained findings were found suitable for carrying out the regression analysis. The regression analysis was carried out in two ways: (1) a regression analysis among each variable of Industry 4.0 and employees' performance, and (2) a regression analysis between Industry 4.0 (a single variable computed by adding the five variables of Industry 4.0) and employees' performance.

TABLE 20.4**Correlations between Industry 4.0 and Employees' Performance**

Correlations	Industry 4.0	Employees' Performance
Industry 4.0	1	
Employees' performance	0.540**	1

Note(s): **Correlation is significant at the 0.01 level (2-tailed).

Source: Author's owner.

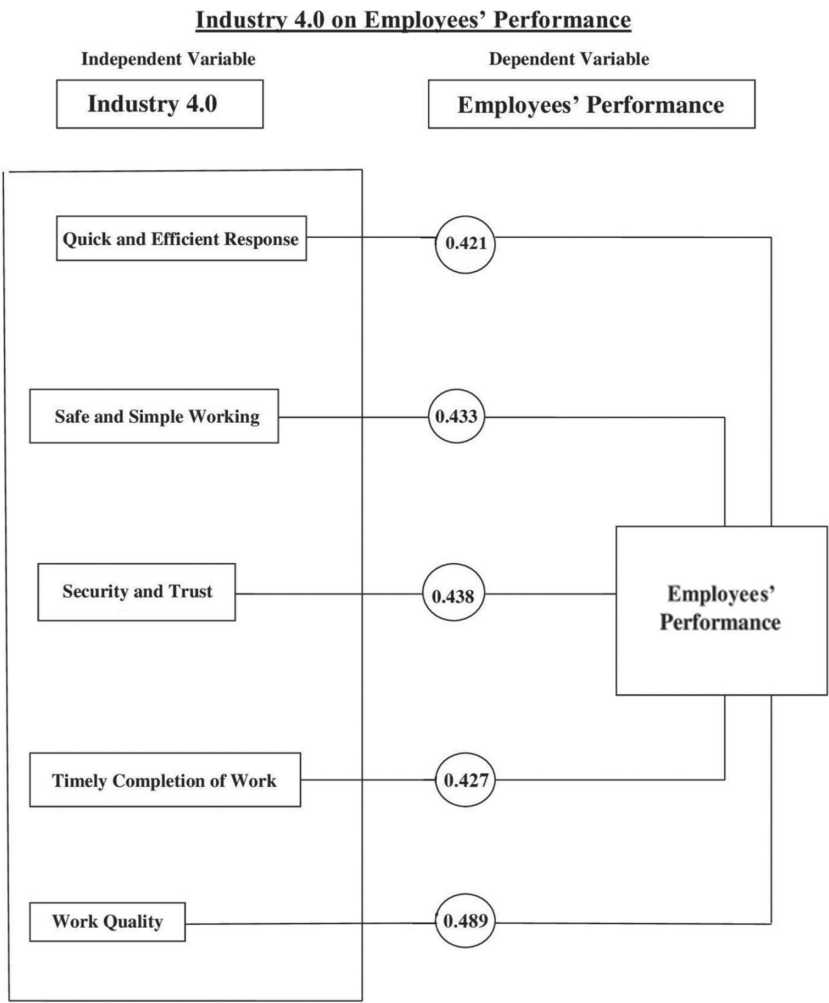


FIGURE 20.2 Research model with obtained correlations.

Source: Author's owner.

For carrying out a regression analysis for all the variables of Industry 4.0 and employees' performance, quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality were taken as the independent variables, and employees' performance was taken as the dependent variable. Table 20.5 shows the results of the regression analysis for the five variables of Industry 4.0 and employees' performance.

It is generally agreed that a hypothesis with a t-value of greater than or equal to 1.96 is considered an accepted hypothesis, whereas a hypothesis with a t-value of less than 1.96 is considered a rejected hypothesis.

TABLE 20.5**Regressions among Five Variables of Industry 4.0 and Employees' Performance**

Regressions (Organizational Performance)	β	t	Sig
Quick & Efficient Response	0.421	10.040	0.000
Safe & Simple Working	0.433	10.375	0.000
Security & Trust	0.438	10.527	0.000
Timely Completion of Work	0.427	10.194	0.000
Work Quality	0.489	12.112	0.000

Source: Author's owner.

Table 20.5 shows that the results yielded a statistically significant relationship between quick and efficient responses and employees' performance. Hypothesis 1 was therefore accepted. Similarly, a statistically significant relationship was obtained between safe and simple working and employees' performance, and thus Hypothesis 2 was not rejected.

Additionally, the relationship between security and trust and employees' performance was statistically significant. Thus, Hypothesis 3 was accepted. Furthermore, the results yielded a statistically significant relationship between the timely completion of work and employees' performance, and therefore Hypothesis 4 was not rejected. It was also found that the relationship between work quality and employees' performance was statistically significant, and Hypothesis 5 was thus accepted.

In addition, the β -values, shown in Table 20.5, indicate that all of the obtained relationships were positive. The hypotheses along with their statistical indicators of relevance are shown in Table 20.6.

In addition, we also carried out the regression analysis between Industry 4.0 (a single variable computed by adding the five variables of Industry 4.0) and employees'

TABLE 20.6**Hypotheses along with Their Statistical Indicators of Relevance**

Statistical Indicators of Relevance				
Hypotheses	T	Correlations	Sig	Result
Hypothesis 1	10.040	0.421	0.000	Accepted
Hypothesis 2	10.375	0.433	0.000	Accepted
Hypothesis 3	10.527	0.438	0.000	Accepted
Hypothesis 4	10.194	0.427	0.000	Accepted
Hypothesis 5	12.112	0.489	0.000	Accepted

Source: Author's owner.

TABLE 20.7
Regressions between Industry 4.0 and Employees’ Performance

Regressions (Employees’ Performance)			
	β	t	Sig
Industry 4.0	0.540	13.877	0.000

Source: Author’s owner.
Table 20.7 shows regressions between Industry 4.0 and employees’ performance.

performance, where Industry 4.0 was taken as the independent variable, and employees’ performance was taken as the dependent variable.

Table 20.7 shows the results of the regression analysis between Industry 4.0 and employees’ performance. Table 20.7 shows that the results also yielded a statistically significant relationship between Industry 4.0 and employees’ performance. Furthermore, the β -value, shown in Table 20.7, indicates that the relationship obtained was positive.

20.5 DISCUSSIONS

This chapter was carried out to assess and determine the impact of Industry 4.0 and five factors of the technology of the era of Industry 4.0, i.e., quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality, on the employees’ performance in the banking sector in India as mentioned above.

The study in this chapter, the findings obtained provided convincing evidence of a strong association among the five factors of the technology and bank employees’ performance. The implementation of Industry 4.0 and its related cutting-edge technologies in the Indian banking sector could help improve employees’ performance.

20.6 CONCLUSION

This study took the unprecedented step to assess and determine the impact of the five factors of the technologies of Industry 4.0 – quick and efficient response, safe and simple working, security and trust, timely completion of work, and work quality – on employees’ performance in the Indian banking sector (Khang et al., 2023b).

The target population was composed of senior-level employees from a few private banks in India – including managers ranging from first-level positions to top-level positions, as well as subordinate employees working under the supervision of first-level managers – who possess the technological know-how of Industry 4.0 (Rani et al., 2023).

The data were collected through a matrix-based survey questionnaire. The findings obtained yielded some interesting findings and provided convincing evidence of a strong association between the five factors of the technologies of Industry 4.0 and the employees' performance in the banking sector (Bhambri et al., 2022).

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